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Suppose n is an integer

Prove: if $n^3 + 5$ is odd, then n is even.

a) Contrapositive: if n is odd, then $n^3 + 5$ is even.

Proof (Contrapositive): Suppose n is odd.

Then $n = 2k + 1$, where k is some integer.

$$\begin{aligned}\text{Then } n^3 + 5 &= (2k + 1)^3 + 5 \\ &= (2k)^3 + (2k)^2(1) + (2k)(1)^2 + 1^3 + 5 \\ &= 8k^3 + 4k^2 + 2k + 6 \\ &= 2(4k^3 + 2k^2 + k + 3)\end{aligned}$$

Since $4k^3 + 2k^2 + k + 3$ is an integer, $n^3 + 5$ is an even integer. □

b) Contradiction:

Proof (Contradiction): Suppose $n^3 + 5$ is odd.

For contradiction, Suppose n is odd.

Then $n = 2k + 1$, where k is some integer.

$$\begin{aligned}\text{Then } n^3 + 5 &= (2k + 1)^3 + 5 \\ &= 2(4k^3 + 2k^2 + k + 3) \quad (\text{Same algebra as part a})\end{aligned}$$

But then this shows that $n^3 + 5$ is even, which contradicts our assumption that $n^3 + 5$ is odd.

Thus, n must be even. □

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23 $P(n)$ = "If a and b are positive real numbers, then $(a+b)^n \geq a^n + b^n$ "

$P(1)$ = "If a and b are positive real numbers, then $a+b \geq a+b$."

Proof (Direct): Suppose a and b are positive real numbers.

Then $a+b = a+b$.

Then $a+b \geq a+b$. □

This is a direct proof.

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29 n is odd if and only if $5n+6$ is odd.

Proof: We will first prove the forward direction directly.

Suppose n is odd. Then $n=2k+1$, where k is an integer.

$$\begin{aligned}\text{Then } 5n+6 &= 5(2k+1) + 6 \\ &= 10k + 5 + 6 \\ &= 10k + 10 + 1 \\ &= 2(5k+5) + 1.\end{aligned}$$

Since $5k+5$ is an integer, $5n+6$ is odd.

Next, we will prove the backwards direction by contraposition.

Suppose n is even. Then $n = 2k$, where k is an integer.

$$\begin{aligned}\text{Then } 5n + 6 &= 5(2k) + 6 \\ &= 10k + 6 \\ &= 2(5k + 3)\end{aligned}$$

Since $5k + 3$ is an integer, $5n + 6$ is even.

(1.7) We will prove this by showing $i \rightarrow ii \rightarrow iii \rightarrow i$

Proof (direct): $i \rightarrow ii$ Suppose $x + 2$ is even.

Then $x + 2 = 2k$ where k is an integer.

$$\begin{aligned}\text{Then } x + 5 &= (x + 2) + 3 \\ &= 2k + 2 + 1 \\ &= 2(k + 1) + 1.\end{aligned}$$

Since $k + 1$ is an integer, $x + 5$ is odd.

Proof (direct): $ii \rightarrow iii$ Suppose $x + 5$ is odd.

Then $x + 5 = 2k + 1$ where k is an integer.

$$\begin{aligned}\text{Then } x^2 &= ((x + 5) - 5)^2 \\ &= (2k + 1 - 5)^2 \\ &= (2k - 4)^2 \\ &= 4k^2 - 16k + 16 \\ &= 2(2k^2 - 8k + 8)\end{aligned}$$

Since $2k^2 - 8k + 8$ is an integer, x^2 is even.

Proof (Contraposition): $iii \rightarrow i$ Suppose $x+2$ is odd.

Then $x+2 = 2k+1$ where k is an integer.

$$\begin{aligned}\text{Then } x^2 &= ((x+2)-2)^2 \\ &= (2k+1-2)^2 \\ &= (2k-1)^2 \\ &= 4k^2 - 4k + 1 \\ &= 2(2k^2 - 2k) + 1\end{aligned}$$

Since $2k^2 - 2k$ is an integer, x^2 is odd.

(1.7)
43 We will prove this by showing $i \rightarrow ii \rightarrow iii \rightarrow iv \rightarrow i$

Proof (Direct): $i \rightarrow ii$ Suppose n is even.

Then $n = 2k$, where k is an integer

Then $n+1 = 2k+1$. Since k is an integer, $n+1$ is odd.

Proof (Direct): $ii \rightarrow iii$ Suppose $n+1$ is odd.

Then $n+1 = 2k+1$ where k is an integer.

$$\begin{aligned}\text{Then } 3n+1 &= 3((n+1)-1) + 1 \\ &= 3(2k+1-1) + 1 \\ &= 3(2k) + 1\end{aligned}$$

Since $2k$ is an integer, $3n+1$ is odd.

Proof (Direct): $iii \rightarrow iv$ Suppose $3n+1$ is odd.

Then $3n+1 = 2k+1$ where k is an integer.

$$\begin{aligned}\text{Then } 3n &= (3n+1) - 1 \\ &= 2k+1 - 1 \\ &= 2k\end{aligned}$$

Since k is an integer, $3n$ is even

Proof (Contraposition): $iv \rightarrow i$ Suppose n is odd. □

Then $n = 2k+1$ where k is an integer.

$$\begin{aligned}\text{Then } 3n &= 3(2k+1) \\ &= 6k+3 \\ &= 6k+2+1 \\ &= 2(3k+1)+1\end{aligned}$$

Since $3k+1$ is an integer, $3n$ is odd. □

(1.8)
5 Proof (Direct): Suppose x and y are real numbers.

Case 1: Suppose $x \geq y$.

$$\begin{aligned}\text{Then } \max(x, y) + \min(x, y) \\ &= x + y\end{aligned}$$

Case 2: Suppose $x < y$.

$$\begin{aligned}\text{Then } \max(x, y) + \min(x, y) \\ &= y + x\end{aligned}$$
□

(1.8)
9 Proof (Direct): Suppose x and y are real numbers.

Case 1: Suppose x and y are both nonnegative.

Then $|x| + |y| = x + y$ and $|x + y| = x + y$, so $|x| + |y| = |x + y|$.

Thus, $|x| + |y| \geq |x + y|$.

Case 2: Suppose x and y are both negative.

Then $|x| + |y| = -x - y$ and $|x + y| = -(x + y) = -x - y$,

so $|x| + |y| = |x + y|$.

Thus $|x| + |y| \geq |x + y|$.

Case 3: Suppose one of x and y is nonnegative and the other is negative.

Without loss of generality, suppose x is nonnegative, y is negative.

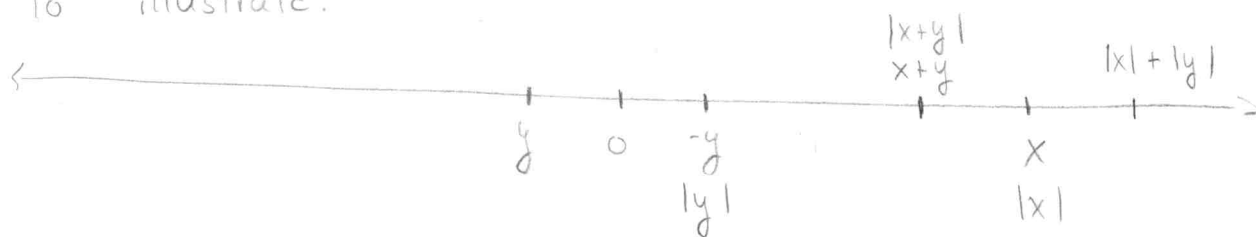
Case 3.1: Suppose $x \geq -y$ (x is "bigger" than y).

Then $x + y \geq 0$, so $|x + y| = x + y < x$.

$|x| + |y| = x + |y| > x$.

Thus, $|x + y| < x < |x| + |y|$, so $|x + y| < |x| + |y|$.

To illustrate:



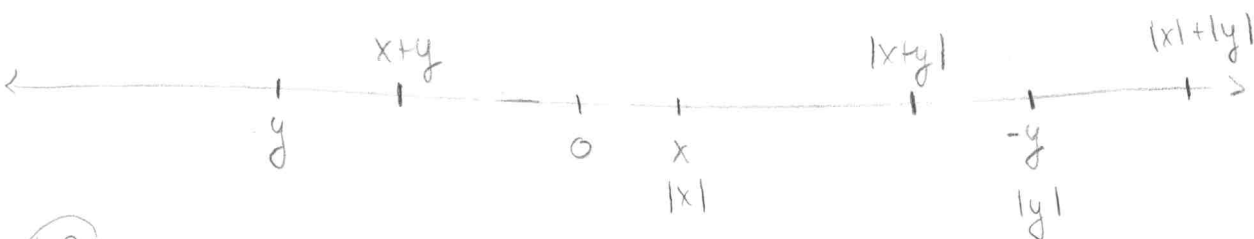
Case 3.2: Suppose $x < -y$ (x is "smaller" than y)

Then $x + y < 0$, so $|x + y| = -(x + y) = (-x) + (-y) \leq -y$

$$|x| + |y| = |x| + (-y) \geq -y.$$

Thus $|x + y| \leq -y \leq |x| + |y|$, so $|x + y| \leq |x| + |y|$.

To illustrate:



1.8
19 Proof (Direct): Suppose a and b are odd integers, $a \neq b$.
We will attempt to find solutions to $|a-c| = |b-c|$.

Case 1: Suppose $(a-c)$ and $(b-c)$ are both nonnegative.

Then $a-c = b-c$ implies $a=b$, which contradicts our assumption that $a \neq b$.

Case 2: Suppose $(a-c)$ and $(b-c)$ are both negative.

Then $|a-c| = |b-c| \Rightarrow -(a-c) = -(b-c) \Rightarrow -a+c = -b+c \Rightarrow a=b$,
a contradiction.

Case 3: Suppose one of $(a-c)$ and $(b-c)$ is nonnegative and the other is negative.

Without loss of generality suppose $(a-c)$ is nonnegative and $(b-c)$ is negative.

$$\text{Then } |a-c| = |b-c|$$

$$\Rightarrow a-c = -(b-c)$$

$$\Rightarrow a-c = -b+c$$

$$\Rightarrow a+b = 2c$$

$$\Rightarrow c = \frac{a+b}{2}$$

Note that since a and b are odd, $a+b$ is even,

so $\frac{a+b}{2}$ is an integer

Thus, $c = \frac{a+b}{2}$ is an integer such that $|a-c| = |b-c|$.

This c is unique because if there existed another integer, say d , such that $|a-d| = |b-d|$, we could solve for d , obtaining $d = \frac{a+b}{2}$, which would be equal to c .

□

(1.8)
(21) Proof (Direct): Suppose n is an odd integer.

We wish to find a k such that $n = (k-2) + (k+3)$

$$\Leftrightarrow n = k-2 + k+3$$

$$\Leftrightarrow n = 2k+1$$

$$\Leftrightarrow \frac{n-1}{2} = k.$$

Since n is odd, $n-1$ is even, so $\frac{n-1}{2}$ is an integer.

Thus $k = \frac{n-1}{2}$ will satisfy $n = (k-2) + (k+3)$.

This k is unique because if there existed another integer, say j , such that $n = (j-2) + (j+3)$, we could solve for j , obtaining $j = \frac{n+1}{2}$ which would be equal to k . □

(1.8)
31 Proof (exhaustive): First, observe that $5^3 = 125 > 100$,
so $n^3 + n^2 = 100$ could only be true for $1 \leq n \leq 4$.

$n=1$: $1^3 + 1^2 = 1 + 1 = 2 \neq 100$

$n=2$: $2^3 + 2^2 = 8 + 4 = 12 \neq 100$

$n=3$: $3^3 + 3^2 = 27 + 9 = 36 \neq 100$

$n=4$: $4^3 + 4^2 = 64 + 16 = 80 \neq 100$.

Thus, there are no positive integers n such that $n^3 + n^2 = 100$. □

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Assume that the upper left and lower right corners are removed.

Note that we can rotate the board so that our numbering matches the suggestion.

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Note that for our board to be tiled, each square must be covered. Namely, square 2 has to be covered.

The domino on square 2 is either vertical or horizontal.

Case 1: Suppose the domino on 2 is vertical (covering 6).

Then we must have dominos on: 5-9, 13-14, 10-11

But then there is no way to cover 15, a contradiction.

So, the domino on 2 can't be vertical.

Case 2: Suppose the domino on 2 is horizontal (covering 3).

Then we must have dominos on: 4-8, 11-12, 6-7, 5-9, 10-14

But then there is no way to cover 15, a contradiction.

Thus, it is not possible to tile the 4×4 checkerboard.